

# **REPOSCAN:Academic Integrity Verification System**

*A Project Report*

*Submitted to the APJ Abdul Kalam Technological University*

*in partial fulfillment of requirements for the award of degree*

***Bachelor of Technology***

*in*

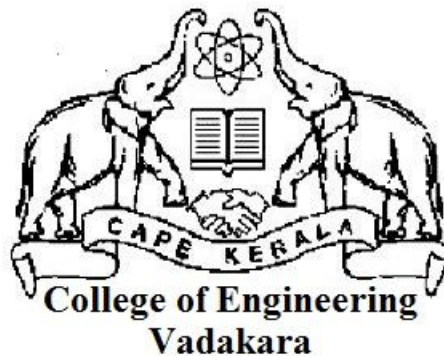
**COMPUTER SCIENCE AND ENGINEERING**

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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**COLLEGE OF ENGINEERING VADAKARA**

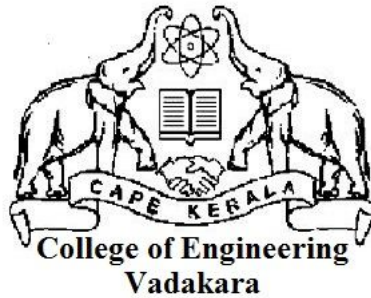
**KERALA**

**APRIL 2026**

**DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING**

**COLLEGE OF ENGINEERING VADAKARA**

**2022 - 26**



**CERTIFICATE**

This is to certify that the report entitled **REPOSCAN:Academic Integrity Verification System** submitted by **Ansa Mariya P S** (VDA22CS014), **Jawahar Sreeram** (VDA22CS034)& **Krishnasai** (VDA22CS037) to the APJ Abdul Kalam Technological University in partial fulfillment of the B.Tech. degree in COMPUTER SCIENCE AND ENGINEERING is a bonafide record of the project work carried out by them under our guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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We hereby declare that the project report **REPOSCAN:Academic Integrity Verification System**, submitted for partial fulfillment of the requirements for the award of degree of Bachelor of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by us under supervision of Prof. Nithya A K

This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources.

We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in my submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

Vadakara  
08-04-2026

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# Abstract

This project presents a robust plagiarism detection system designed to uphold academic integrity through a sophisticated hybrid similarity approach. Moving beyond traditional string matching, the system implements a multi-layered pipeline that balances computational efficiency with deep semantic analysis. The process begins with secure document ingestion, where a SHA-256 hashing mechanism generates unique digital fingerprints to instantly identify exact or near-duplicate submissions within institutional repositories. For advanced detection of paraphrased content, the system employs an NLP-based preprocessing pipeline including tokenization, stop-word removal, and lemmatization to standardize textual data for precise comparison.

Central to the architecture is the use of Transformer-based models to convert extracted text into high-dimensional vector representations that capture contextual meaning. These embeddings are indexed and managed using FAISS (Facebook AI Similarity Search), enabling high speed, large-scale similarity retrieval across extensive document corpora. The detection engine integrates these methods by calculating Cosine Similarity to evaluate semantic overlap, ensuring the identification of disguised plagiarism that eludes lexical tools. The system automates the generation of comprehensive reports, providing document-wide similarity scores and detailed source attribution to facilitate faculty review and foster ethical writing practices.

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# Chapter 1

## Introduction

Plagiarism has become a pressing concern in academic, professional, and digital content creation fields, where the ease of accessing and copying information threatens the originality and integrity of work. It encompasses a variety of deceptive behaviors, from direct copying of text to subtler forms like paraphrasing without proper citation. The rapid growth of digital academic content and online publications further complicates the task of identifying copied material manually. As a result, reliable and efficient plagiarism detection mechanisms are indispensable for preserving intellectual property rights, maintaining ethical standards, and ensuring trust in scholarly communication.

This project proposes a comprehensive plagiarism detection system that utilizes a hybrid similarity approach to address these challenges. The system architecture follows a modular design to efficiently detect duplication using both cryptographic and semantic methods. The process begins with secure document ingestion, where a SHA-256 hashing mechanism generates unique digital fingerprints to instantly identify exact or near-duplicate submissions within institutional repositories. For advanced detection of paraphrased content, the system employs an NLP processing pipeline including tokenization, stop-word removal, and lemmatization to standardize textual data for precise comparison.

Central to the architecture is the use of transformer-based models to convert extracted text into high-dimensional vector representations that capture contextual meaning. These embeddings are indexed and managed using FAISS (Facebook AI Similarity Search), enabling high-speed, large-scale similarity retrieval across

extensive document corpora. The detection engine integrates these methods by calculating cosine similarity to evaluate semantic overlap, ensuring the identification of disguised plagiarism that eludes traditional lexical or string-matching tools. A key feature is the document-wide analysis, which computes a total plagiarism score by evaluating the proportion of matched text against the entire extracted content.

In addition to detection accuracy, the project places strong emphasis on scalability, data privacy, and user-centric design. The system automates the generation of comprehensive reports, providing document-wide similarity scores, highlighted matches, and detailed source attribution to facilitate faculty review. The modular design supports future extensions, enabling adaptation to emerging AI techniques to further fortify resistance against sophisticated plagiarism tactics. By integrating these advanced techniques, the solution aims to serve as an essential tool to uphold academic integrity, streamline verification workflows, and foster ethical writing practices in an increasingly digital world.

# Chapter 2

## Literature Review

### 2.1 Large Language Model-Based Plagiarism Detection System for Academic Journals

*Feiyang Liu and Kewei You, "Application of academic misconduct detection system and suggestions for Chinese scientific journals," Chinese Journal of Scientific and Technical Periodicals, 2025*

The research introduces an innovative framework utilizing Large Language Models to identify semantic plagiarism across scholarly manuscripts, addressing the limitations of traditional string-matching tools. Instead of depending solely on lexical or keyword overlap, the model applies contextual embeddings through transformer architectures to capture the underlying meaning and intent behind sentences. This approach allows the system to detect implicit, paraphrased, and even cross-lingual plagiarism with significantly higher accuracy and efficiency. The architecture is composed of a multi-stage pipeline beginning with a retrieval engine that searches across extensive journal databases to identify potential source documents.

Once potential matches are found, a classifier is employed to distinguish between legitimate original writing and heavily paraphrased content. To aid in academic review, the system provides detailed similarity visualization, which highlights specific copied regions and provides numerical semantic overlap scores. This system emphasizes semantic comprehension over direct string comparison, ensuring the accurate detection of AI-generated content or sophisticated paraphrasing that often escapes traditional

detection tools. By adopting this embedding-based method, the framework supports institutional repositories in maintaining academic integrity by detecting idea-level plagiarism within project reports and scholarly papers. Ultimately, the integration of these advanced techniques helps reduce the manual effort required for verification while fortifying the system's resistance against sophisticated academic misconduct.

## **2.2 Automated Plagiarism Detection in Moodle**

*Ravi Kishore Kodali, Tanvi Shekhar, and Lakshmi Boppana, "Automated Plagiarism Detection in Handwritten Documents using OCR and AWS Services," 2023.*

This study focuses on an automated plagiarism detection system integrated within the Moodle learning platform to streamline academic integrity workflows. The proposed model automates every stage of the detection process, beginning with document submission and moving through text preprocessing, similarity calculation, and final report generation. By utilizing term frequency-inverse document frequency (TF-IDF) and cosine similarity, the system compares new student uploads with previously stored materials and external databases. The research highlights a modular workflow that allows for the batch processing of numerous documents, providing a significant scalability advantage in large academic settings. Additionally, the authors emphasize maintaining data privacy by ensuring all submissions are processed within the institutional network.

The integration with the Moodle learning management system (LMS) allows for a seamless user experience where student reports are automatically moved into a processing pipeline. Once the document is parsed, the system extracts the text and performs cleaning operations such as stop-word removal, lemmatization, and normalization. These processed text files are stored in a dedicated repository for internal comparison against future submissions. This automated approach demonstrates the effectiveness of combining preprocessing, indexing, and efficient vector-based analysis for large datasets, a principle central to ensuring academic honesty. Following the evaluation, the system compiles similarity data and percentage scores into a detailed report for faculty review

## 2.3 Code Plagiarism Detection Function in C Programming Course

*Xiqin Lu and Khaing Hsu Wai studied The Application of Code Plagiarism Detection Functions in university-level C programming courses.*

This system focuses on identifying structural similarities within programming assignments using multi-stage preprocessing and syntactic analysis. After normalizing code by removing comments, renaming identifiers, and formatting syntax, the system employs token-based analysis and Abstract Syntax Tree comparison to measure logical equivalence between different code submissions. To enhance detection precision, the authors implemented Levenshtein distance to compute structural similarity between functions, effectively distinguishing legitimate code reuse from intentional copying. The experimental results proved the model's efficiency in identifying logical similarities that standard text-matching tools often miss.

This methodology is highly relevant for technical reports that contain algorithmic or code components. By adapting the normalization and Abstract Syntax Tree-based analysis, a detection system can extend its capabilities beyond plain text to include programmatic content and logical structures. This hybrid ability ensures that plagiarism within code snippets embedded in academic reports can be detected effectively. Such a multi-tiered approach allows for the identification of logically identical work even when variables and comments have been significantly altered by the user.

## 2.4 Faculty Perspectives on Academic Integrity and Student Cheating

*Rebecca Strachan, Emma Anderson, and Cynthia Oguna, "An Exploratory Study on Academic Integrity and Student Cheating among senior computing faculty," 2025.* This exploratory study investigates educators' perceptions of academic misconduct and their level of trust in existing technological detection tools. The research reveals that while automated systems are being widely adopted, many faculty members express

concerns regarding the interpretability and fairness of the generated results. Educators stress that detection systems should produce explainable, evidence-based reports that not only identify copied content but also serve a clear educational purpose for students. The authors highlight the growing challenge posed by AI-generated text and recommend the development of pedagogically oriented tools that foster responsibility and awareness among learners, ensuring that detection is combined with transparency and constructive feedback.

The findings from this research emphasize that a purely punitive approach to plagiarism detection is often insufficient for long-term academic integrity. Instead, there is a strong call for systems that prioritize transparency, allowing students to see exactly how their work is being analyzed and where similarities have been flagged. By providing evidence-based results and clear source attributions, the framework aims to combine detection with education, helping students understand the boundaries of original work. This approach ensures that the verification process is viewed as a constructive tool for improving writing responsibility rather than a purely technical barrier.

Furthermore, the study points toward the need for administrative dashboards that allow faculty to interpret results within a broader context. Such tools should provide detailed visualizations and section-wise breakdowns—or in some cases, document-wide analysis—to help reviewers make informed decisions between unintentional citation errors and deliberate misconduct. By fostering this collaborative environment between technology and pedagogy, educational institutions can better support research originality and promote ethical content creation practices across diverse professional and academic landscapes.

## **2.5 AI-Driven Plagiarism Detection and Hybrid Similarity Models**

*Aditya Raj and Meghna Sharma, "A Survey on AI-Driven Plagiarism Detection and Hybrid Similarity Models," 2024.* This extensive survey analyzes modern techniques that integrate natural language processing, deep learning, and stylometric analysis to

combat academic dishonesty. The research traces the evolution of detection from traditional keyword-based comparisons to hybrid neural frameworks that combine statistical methods like n-grams with semantic embeddings and stylistic writing style analysis. The authors emphasize that these hybrid models deliver significantly higher accuracy when detecting paraphrased or AI-generated content that typically eludes simpler lexical systems. By incorporating stylometric profiling, the system can differentiate unique writing patterns, ensuring a more comprehensive and fair evaluation of document originality.

The study also discusses the challenges of implementing such complex systems, particularly regarding computational overhead and the need for explainable outputs. Integrating transformer-based embeddings with traditional statistical features allows for a more robust detection pipeline that can identify idea level plagiarism. This approach is particularly effective in identifying sophisticated academic misconduct where the structure or wording has been altered but the core content remains copied. Ultimately, the survey highlights that a multi tiered analysis—combining semantic, statistical, and stylistic features is necessary to provide a reliable and fair plagiarism evaluation in the modern digital landscape.



# **Chapter 3**

## **System Architecture and Data Flow Design**

### **3.1 Introduction**

The proposed system, REPOSCAN – Academic Integrity Verification System, is developed to verify the originality of academic documents such as project reports, seminar papers, and theses submitted by students. Its main objective is to maintain academic integrity by identifying similarities between newly uploaded documents and those stored in the existing repository. This process minimizes plagiarism and ensures that every academic submission is unique and authentic. The system automates the tasks of document uploading, text extraction, comparison, and report generation, thus reducing manual effort while improving efficiency and accuracy.

The architecture of REPOSCAN has been designed to be modular, scalable, and easy to maintain. It divides the system into multiple layers and modules, each responsible for a specific function. The following sections describe the system architecture, its components, internal working, and data flow representation through DFD diagrams.

## 3.2 System Architecture

The REPOSCAN system architecture is redesigned as a hybrid, scalable multi-layer architecture that integrates hashing techniques, Natural Language Processing (NLP), and vector-based similarity search. The system follows a three-tier model consisting of the User Interface Layer, Application Layer, and Database Layer, with an enhanced processing pipeline to support large-scale academic document analysis.

Unlike traditional plagiarism detection systems, REPOSCAN incorporates a dual detection mechanism combining hash-based exact matching and FAISS-based semantic similarity search, ensuring both speed and accuracy. The architecture is designed to efficiently handle large institutional repositories while supporting real-time document analysis.

The workflow begins with document upload and proceeds through text extraction, preprocessing, hashing, vector generation, similarity search, and report generation, forming a complete automated pipeline. This modular structure ensures scalability, faster processing, and improved detection of both exact and paraphrased plagiarism.

### 3.2.1 Core Components

#### User Interface Layer

The User Interface Layer acts as the interaction point between users and the system. It allows students and faculty members to register and log in securely, upload academic documents (PDF format), and view plagiarism reports and similarity scores.

The interface is developed using web technologies and ensures usability, accessibility, and secure authentication. It also provides visualization of results, including similarity percentages and highlighted matched content.

#### Application Layer

The Application Layer is the core of the REPOSCAN system and is responsible for executing all major processing tasks. It consists of multiple integrated modules forming a hybrid detection pipeline.

**Document Upload and Validation Module** This module handles file uploads and ensures that only valid academic documents (PDF format) are accepted. Metadata such as file name and user details are stored securely.

**Text Extraction Module** This module extracts textual content from uploaded documents using tools such as PyMuPDF. It focuses on extracting meaningful academic content while ignoring irrelevant sections.

**Text Preprocessing Module** This module prepares extracted text for analysis by applying tokenization, stop-word removal, lemmatization, and noise removal. These steps ensure uniform and clean data for accurate similarity detection.

**Hashing Module** This module uses hashing techniques such as SHA-256 through the hashlib library to generate unique fingerprints for documents. It enables fast detection of exact or near-duplicate documents and reduces computational overhead.

**Vector Generation Module** The preprocessed text is converted into numerical representations using NLP-based embedding techniques such as Transformer models. This helps capture semantic meaning and detect paraphrased content.

**Similarity Detection Module** This module uses FAISS (Facebook AI Similarity Search) to perform efficient large-scale similarity comparison. It retrieves similar documents from the repository using vector search and computes similarity using cosine similarity.

**Plagiarism Scoring and Report Generation Module** This module computes the plagiarism percentage using the formula:

$$P = \frac{\text{Matched Text}}{\text{Total Extracted Text}} \times 100$$

It generates a detailed report including similarity scores, matched sections, source references, and classification levels such as low, medium, and high plagiarism.

## **Database Layer**

The Database Layer acts as the central storage unit of the system. It stores user credentials, uploaded documents, extracted text, hash values, vector embeddings, and generated reports.

It also serves as the repository for all academic documents used during similarity analysis. The database ensures secure storage, efficient retrieval, and supports FAISS indexing for fast similarity search. This enables repository-based plagiarism detection, which improves accuracy compared to traditional web-based systems.

## **Working of the System**

The overall working of REPOSCAN begins with user authentication, where users securely log in to access the system. Once authenticated, the document upload module allows users to submit academic files in PDF format, which are then securely stored and forwarded to the processing pipeline. The text extraction module retrieves the content from the uploaded document using tools such as PyMuPDF, focusing only on the core academic sections from the introduction to the conclusion while eliminating irrelevant parts like certificates, acknowledgements, and annexures using NLP techniques and pattern-based filtering. The extracted text is then cleaned and preprocessed through tokenization, stop-word removal, lemmatization, and noise reduction to ensure consistency and accuracy. The system then performs a dual-stage similarity detection process, where the hashing module first generates a unique fingerprint of the document using SHA-256 via the hashlib library to quickly detect exact or near-duplicate documents. Subsequently, the preprocessed text is transformed into numerical vector representations using NLP-based embedding techniques such as Transformer models to capture semantic meaning. These vectors are then compared against repository data using FAISS, which enables efficient large-scale similarity search, and cosine similarity is applied to determine the degree of similarity between documents. Based on the detected matches, the system calculates similarity scores and identifies corresponding source documents. Finally, the report generation module compiles the results into a detailed plagiarism report containing similarity percentages, highlighted matched sections, and source references, which is stored in the database.

and made accessible to the user through the web interface.

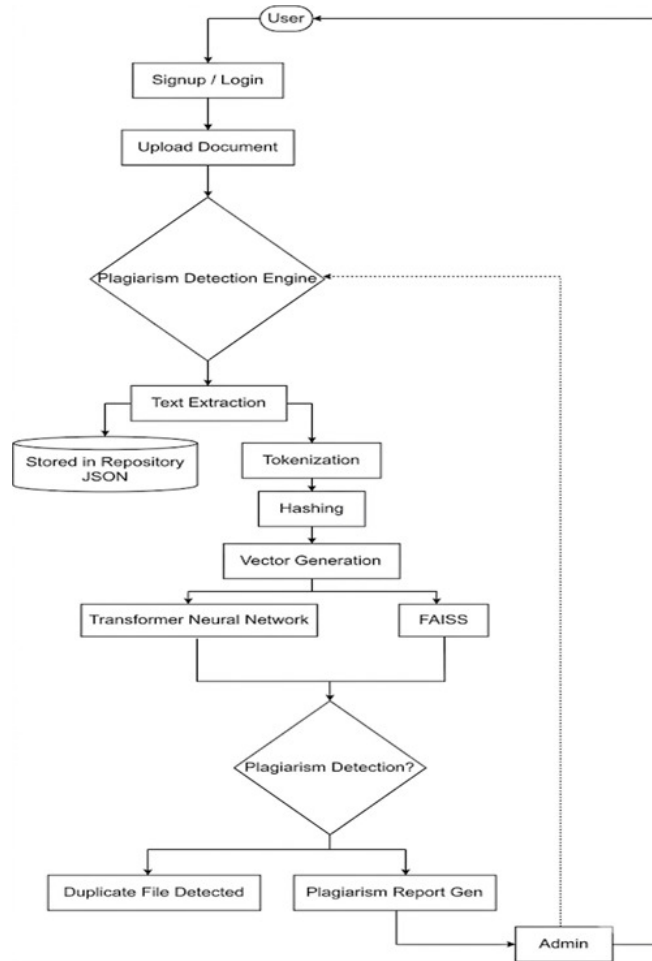


Figure 3.1: System Flowchart

### 3.3 Data Flow Diagram

The Data Flow Diagram (DFD) represents how data moves through the REPOSCAN system and illustrates the interaction between users, system processes, and data storage. It provides a clear understanding of how input data is transformed into meaningful output through different stages of processing.

#### 3.3.1 Level 0 Data Flow Diagram

At Level 0, the REPOSCAN system is represented as a single high-level process that interacts with external entities such as the user and the database. The process begins when the user uploads an academic document through the system interface. The

uploaded document is then processed by the REPOSCAN system, where operations such as text extraction, preprocessing, hashing, and vector-based similarity analysis are performed. The system retrieves stored documents, hash values, and vector embeddings from the database to carry out similarity comparison using FAISS. After completing the analysis, a detailed plagiarism report containing similarity scores and matched content is generated and returned to the user. This Level 0 diagram provides an overall view of the system workflow, highlighting the input, processing, storage, and output interactions.

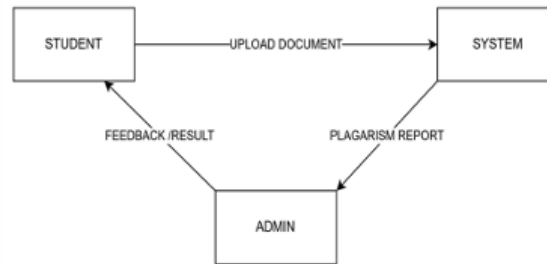


Figure 3.2: Level 0 DFD

### 3.3.2 Level 1 Data Flow Diagram

In the Level 1 DFD, the REPOSCAN system is decomposed into its core subprocesses to provide a detailed view of internal operations. These subprocesses include User Authentication, Document Upload, Text Extraction and Preprocessing, Hashing and Vector Generation, Similarity Detection, and Report Generation. The process begins when a user logs into the system through the User Authentication module. After successful authentication, the user uploads an academic document, which is stored temporarily and passed to the Text Extraction and Preprocessing module.

The extracted content is cleaned and normalized using NLP techniques to prepare it for analysis. The processed text is then forwarded to the Hashing and Vector Generation module, where a unique hash value is generated using SHA-256 for exact match detection, and vector embeddings are created using NLP-based models to capture semantic meaning. These outputs are then passed to the Similarity Detection module, where FAISS is used to perform efficient similarity search by comparing the input document with stored repository data. The system retrieves relevant document data, hash values, and embeddings from the database during this process. Finally,

the Report Generation module compiles the similarity results, matched content, and plagiarism scores into a structured report, which is stored in the database and presented to the user.

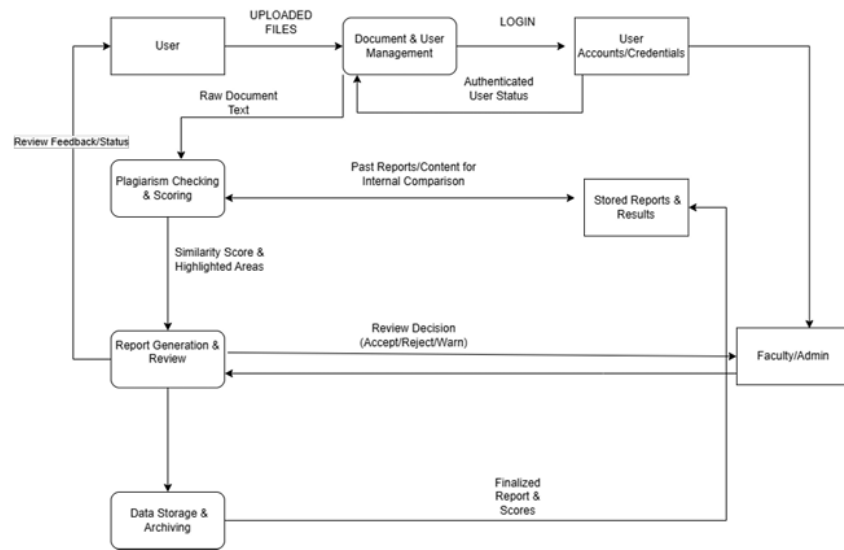


Figure 3.3: Level 1.1 DFD

## 3.4 Module Description

The REPOSCAN system is composed of multiple interconnected modules that collaboratively perform the task of academic document verification. Each module is designed to handle a specific stage in the processing pipeline, ensuring a structured and efficient workflow. The modular approach improves system scalability, maintainability, and performance, while enabling the integration of advanced techniques such as hashing, Natural Language Processing (NLP), and vector-based similarity search for accurate plagiarism detection.

### 3.4.1 User Authentication Module

The User Authentication Module is responsible for managing secure access to the REPOSCAN system. It ensures that only authorized users such as students and faculty members can interact with the system. When a user attempts to log in, their credentials are verified against the stored records in the database. To enhance security, passwords are not stored in plain text but are encrypted using secure hashing techniques.

The module also supports user registration, allowing new users to create accounts by providing necessary details. Input validation mechanisms are applied to prevent invalid or malicious entries. Upon successful authentication, a session is created to maintain user state throughout the interaction. If authentication fails, the system provides appropriate feedback while preventing unauthorized access attempts. This module plays a critical role in protecting sensitive academic data and ensuring controlled system usage.

#### **Algorithm: User Authentication Module**

1. User submits login or registration request.
2. If registration:
  - (a) Validate input fields (email, password, etc.).
  - (b) Encrypt password using hashing algorithm.
  - (c) Store user details in database.
3. If login:
  - (a) Retrieve stored credentials from database.
  - (b) Hash input password and compare with stored hash.
  - (c) If match, grant access and create session.
  - (d) Else, deny access and show error message.
4. Maintain session for authenticated user.
5. Logout clears session data.

#### **3.4.2 Document Upload Module**

The Document Upload Module facilitates the submission of academic documents by authenticated users. It primarily supports PDF format to ensure consistency in processing and extraction. When a file is uploaded, the module performs validation checks such as file type verification, size constraints, and format compliance to prevent errors during processing.



Once validated, the document is stored temporarily in the repository along with associated metadata such as user ID, upload timestamp, and document name. The module also ensures secure handling of files to prevent data corruption or unauthorized access. Additionally, it acts as a trigger point for the processing pipeline by forwarding the uploaded document to the text extraction module. This module ensures reliable and efficient input handling for the entire system.

**Algorithm: Document Upload Module**

1. Authenticated user selects a file to upload.
2. System checks file format (must be PDF).
3. Validate file size and integrity.
4. If validation fails, reject upload.
5. If validation succeeds:
  - (a) Store file temporarily in repository.
  - (b) Record metadata (user ID, timestamp, filename).
6. Forward document to text extraction module.

### **3.4.3 Text Extraction and Filtering Module**

The Text Extraction and Filtering Module is responsible for extracting meaningful textual content from uploaded documents. It uses specialized libraries such as PyMuPDF to read and extract text from PDF files. The module focuses on identifying and isolating the main academic content by detecting structural markers such as “Introduction,” “Chapter 1,” and “Conclusion.”

To improve the quality of extracted data, the module applies filtering techniques to remove irrelevant sections such as title pages, certificates, acknowledgements, references, and annexures. This is achieved using a combination of regular expressions and NLP-based pattern recognition. By isolating only the core content, the system ensures that similarity analysis is performed on relevant information, thereby improving accuracy and reducing noise in the detection process.

#### **Algorithm: Text Extraction and Filtering Module**

1. Receive uploaded PDF document.
2. Extract text using PyMuPDF.
3. Identify structural sections (Introduction to Conclusion).
4. Remove unwanted sections (title, certificate, references, etc.).
5. Apply regex and NLP filtering techniques.
6. Output cleaned core academic content.

#### **3.4.4 Text Preprocessing Module**

The Text Preprocessing Module prepares the extracted text for further analysis by cleaning and standardizing it. This module removes unwanted elements such as punctuation, special characters, digits, and stop words that do not contribute to the meaning of the text. It then performs tokenization, which breaks the text into smaller units such as words or tokens.

Further processing includes lemmatization, which reduces words to their base or root form, ensuring that different variations of the same word are treated uniformly. Advanced NLP libraries such as NLTK and SpaCy are used to perform these operations efficiently. The preprocessing stage significantly enhances the quality of data by reducing redundancy and improving consistency, which is essential for accurate similarity detection. The cleaned text is then stored for use in the next stage of processing.

#### **Algorithm: Text Preprocessing Module**

1. Input extracted text.
2. Remove punctuation, digits, and special characters.
3. Remove stop words.
4. Perform tokenization.

5. Apply lemmatization using NLP libraries.
6. Normalize text data.
7. Store processed text for next stage.

### **3.4.5 Hashing and Vector Generation Module**

The Hashing and Vector Generation Module is a key component that enables both exact and semantic similarity detection. In the first stage, the module uses the hashlib library to generate a SHA-256 hash value for each document. This hash acts as a unique fingerprint, allowing the system to quickly identify exact or near-duplicate documents without performing complex computations.

In the second stage, the module converts the preprocessed text into numerical vector representations using NLP-based embedding techniques such as Transformer models. These embeddings capture the contextual and semantic relationships between words, enabling the system to detect paraphrased or conceptually similar content. The combination of hashing and vector generation provides a hybrid approach that balances speed and accuracy in plagiarism detection.

#### **Algorithm: Hashing and Vector Generation Module**

1. Input preprocessed text.
2. Generate SHA-256 hash using hashlib.
3. Store hash value for duplicate detection.
4. Convert text into embeddings using NLP model.
5. Generate numerical vectors representing text semantics.
6. Store vectors for similarity comparison.

### **3.4.6 Similarity Detection Module**

The Similarity Detection Module is the core analytical component of the REPOSCAN system. It is responsible for comparing the processed document with existing

documents stored in the repository. The module utilizes FAISS (Facebook AI Similarity Search), which is an efficient library designed for large-scale vector similarity search.

Using FAISS, the system performs high-speed comparison of document embeddings to identify similar content across a large dataset. The module retrieves the most relevant matches based on similarity scores and ranks them accordingly. In addition to vector-based comparison, the module also checks hash values to detect exact duplicates quickly. This dual approach ensures that both direct copying and paraphrased plagiarism are effectively identified. The module is optimized for scalability, allowing it to handle large volumes of academic documents efficiently.

#### **Algorithm: Similarity Detection Module**

1. Input document hash and vectors.
2. Compare hash with database hashes.
3. If match found:
  - (a) Mark as exact duplicate.
4. Else:
  - (a) Perform vector comparison using FAISS.
  - (b) Retrieve top-k similar documents.
  - (c) Compute similarity scores.
5. Rank results based on similarity.
6. Forward results to report generation module.

#### **3.4.7 Report Generation Module**

The Report Generation Module is responsible for presenting the results of the similarity analysis in a clear and structured format. It compiles information such as overall similarity percentage, matched text segments, and references to source

documents. The module highlights plagiarized sections within the document, making it easier for users to identify problematic areas.

In addition to textual information, the report may include visual indicators such as color-coded highlights to enhance readability. The module also categorizes the level of plagiarism into different levels such as low, medium, and high based on predefined thresholds. Once generated, the report is securely stored in the database and made accessible to users through the system interface. This module ensures transparency, helps users understand the results, and provides actionable insights to improve document originality.

#### **Algorithm: Report Generation Module**

1. Receive similarity results.
2. Calculate overall similarity percentage.
3. Identify and highlight matched text segments.
4. Retrieve corresponding source documents.
5. Classify plagiarism level (low, medium, high).
6. Generate structured report.
7. Store report in database.
8. Display report to user.

### **3.5 System Requirements**

The development and deployment of the REPOSCAN – Academic Integrity Verification System require both hardware and software components capable of supporting advanced text processing, hashing operations, and large-scale similarity search. The system is designed to operate efficiently in academic environments while ensuring scalability, performance, and reliability. The following specifications outline the minimum and recommended requirements for proper system execution.

### 3.5.1 Hardware Requirements

The hardware requirements are defined to ensure smooth execution of NLP processing, hashing operations, and FAISS-based similarity search.

- **Processor:** Intel Core i5 or higher (multi-core processor recommended) to efficiently handle text preprocessing, hashing, and vector-based similarity computations.
- **RAM:** Minimum 8 GB RAM (16 GB recommended) to support large document processing, embedding generation, and fast similarity search using FAISS.
- **Storage:** At least 500 GB hard disk space for storing uploaded documents, hash values, vector embeddings, and generated reports. Higher storage (1 TB or more) is recommended for large institutional repositories.
- **Monitor:** Standard HD display with minimum resolution of 1366×768 pixels for clear visualization of the user interface and plagiarism reports.
- **Network:** Stable internet or intranet connectivity for document upload, server communication, and database access.
- **Input Devices:** Standard keyboard and mouse for user interaction.
- **Server (Multi-user Deployment):** A dedicated server with multi-core processor, minimum 16 GB RAM, and high storage capacity to support concurrent users and large-scale FAISS indexing operations.

### 3.5.2 Software Requirements

The software requirements define the tools, frameworks, and libraries used to implement the REPOSCAN system, ensuring efficient processing and accurate plagiarism detection.

- **Operating System:** Windows 10 or later, Ubuntu Linux, or macOS for development and deployment environments.

- **Programming Language:** Python 3.8 or higher for implementing backend logic, NLP processing, hashing, and similarity detection algorithms.
- **Framework:** Flask or Django framework for developing the web-based user interface and handling server-side operations.
- **Text Extraction Libraries:**
  - PyMuPDF – for extracting text from PDF documents.
- **Natural Language Processing Libraries:**
  - NLTK and spaCy – for tokenization, stop-word removal, and lemmatization.
  - Transformer-based models – for generating semantic embeddings of text.
- **Hashing and Similarity Libraries:**
  - hashlib – for generating SHA-256 hash values for document fingerprinting.
  - FAISS (Facebook AI Similarity Search) – for efficient large-scale vector similarity search.
  - NumPy and pandas – for numerical computations and data handling.
- **Database:** MySQL or PostgreSQL for storing user credentials, document metadata, hash values, embeddings, and plagiarism reports.
- **Frontend Technologies:** HTML, CSS, and JavaScript for designing the user interface and visualizing plagiarism reports.
- **Web Server:** Flask development server, XAMPP (local server), or cloud platforms such as AWS or Heroku for deployment.
- **IDE / Code Editor:** Visual Studio Code or PyCharm for development, debugging, and testing.
- **Browser:** Google Chrome, Mozilla Firefox, or Microsoft Edge for accessing the REPOSCAN web interface.

The above configuration ensures that the REPOSCAN system operates efficiently by supporting fast hashing, accurate semantic analysis, and scalable similarity search. The integration of FAISS and NLP-based models enhances performance when handling large academic repositories, making the system suitable for institutional deployment and future expansion.



# Chapter 4

## Result

### 4.1 Introduction

This chapter presents the experimental results and performance evaluation of the REPOSCAN system, developed for academic integrity verification and plagiarism detection. The effectiveness of the proposed system is analyzed using various test cases involving both textual documents and source code samples. The evaluation focuses on key performance metrics such as accuracy, response time, and similarity detection efficiency. In addition, the system's ability to identify different levels of plagiarism, including exact matches, partial modifications, and completely unrelated content, is examined.

The integration of hashing techniques using hashlib and efficient similarity search through FAISS is also assessed to determine improvements in speed and scalability compared to traditional methods. The results obtained demonstrate the capability of the system to provide reliable and efficient plagiarism detection suitable for academic environments.

#### 4.1.1 Experimental Setup

The experimental setup describes the environment and tools used to evaluate the performance of the REPOSCAN system. The system was developed and tested on a standard computing platform with an Intel i5 processor and 8GB RAM, ensuring compatibility with commonly available academic infrastructure.

The implementation was carried out using Python along with the Flask framework for building the web interface. The hashlib library was used for generating hash values of documents, enabling efficient data handling, while FAISS (Facebook AI Similarity Search) was utilized for fast and scalable similarity search across large datasets.

A dataset consisting of academic text documents and source code files was used for testing. The dataset included original files, partially modified content, and completely unrelated documents to evaluate the system's ability to detect different levels of plagiarism.

All experiments were conducted in a local environment, and the system performance was measured based on similarity scores, detection accuracy, and response time.

#### **4.1.2 Test Cases and Input Data**

To evaluate the performance of the REPOSCAN system, a set of test cases was designed using academic documents collected from the college repository, including seminar reports and project reports. The objective of these test cases is to analyze the system's ability to detect different levels of textual plagiarism effectively.

The dataset used for testing consists of the following categories:

**1. Original Documents:** These documents contain completely unique content with no similarity to other reports in the dataset. This category is used to ensure that the system correctly identifies non-plagiarized documents and produces low similarity scores.

**2. Highly Similar Documents:** This category includes documents that are exact copies or contain very minor changes such as formatting differences or small word modifications. These test cases help evaluate the system's ability to detect direct plagiarism with high similarity scores.

**3. Partially Modified Documents:** These documents include paraphrased content, sentence restructuring, or slight modifications in wording. This category is used to test the system's effectiveness in identifying partial or indirect plagiarism.

**4. Completely Different Documents:** These documents are unrelated in topic and content. This test case ensures that the system produces very low similarity scores for non-related documents, thereby minimizing false positives.

All documents were processed using hashing techniques and FAISS-based similarity search. The generated similarity scores from these test cases are used to evaluate the overall performance, accuracy, and efficiency of the REPOSCAN system.

### 4.1.3 Results and Performance Analysis

This section presents a detailed evaluation of the REPOSCAN system based on various performance metrics. The system was tested using academic documents such as seminar and project reports to measure its effectiveness in detecting plagiarism.

#### Similarity Detection Results

Table ?? shows the similarity scores obtained for different document pairs.

| Test Case       | Similarity Score (%) | Detection Result    |
|-----------------|----------------------|---------------------|
| Document A vs B | 90%                  | High Plagiarism     |
| Document C vs D | 70%                  | Moderate Plagiarism |
| Document E vs F | 30%                  | Low Plagiarism      |
| Document G vs H | 12%                  | No Plagiarism       |

Table 4.1: Plagiarism Detection Results

#### Graphical Comparison with Existing Systems

This section presents a comparative analysis of the REPOSCAN system with existing plagiarism detection approaches based on standard performance metrics such as Accuracy, Precision, Recall, and F1-Score. The comparison is derived from insights obtained through the literature survey, including traditional methods, hybrid machine learning models, and deep learning-based systems.

**Accuracy Comparison** Accuracy represents the overall correctness of a plagiarism detection system in identifying both plagiarized and non-plagiarized documents.

Traditional systems based on keyword matching and TF-IDF provide moderate accuracy due to their inability to capture semantic meaning. Hybrid machine learning models improve performance by combining statistical and semantic features. Deep learning models achieve the highest accuracy due to their contextual understanding of text.

The REPOSCAN system achieves high accuracy by integrating hashing for exact match detection and FAISS-based semantic similarity search, ensuring reliable and efficient performance.

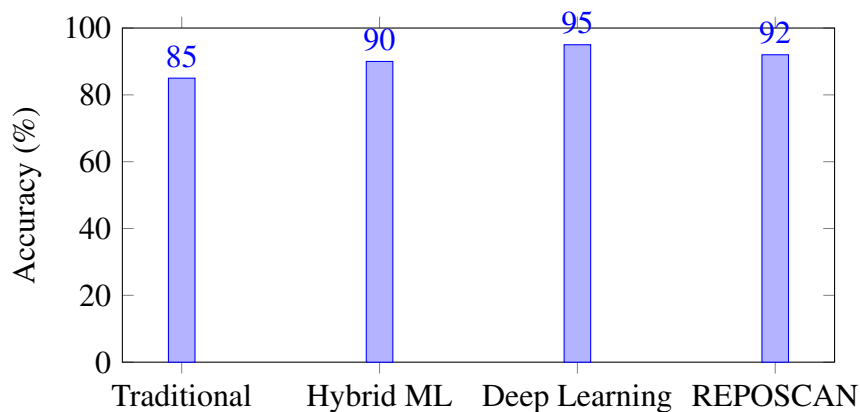


Figure 4.1: Accuracy Comparison with Existing Systems

**Precision Comparison** Precision measures how accurately the system identifies plagiarism without generating false positives.

Traditional systems often show lower precision due to surface-level matching. Hybrid models improve precision using combined techniques, while deep learning models achieve very high precision through semantic understanding.

The REPOSCAN system maintains strong precision by combining SHA-256 hashing and FAISS-based similarity, reducing false positives and ensuring reliable detection.

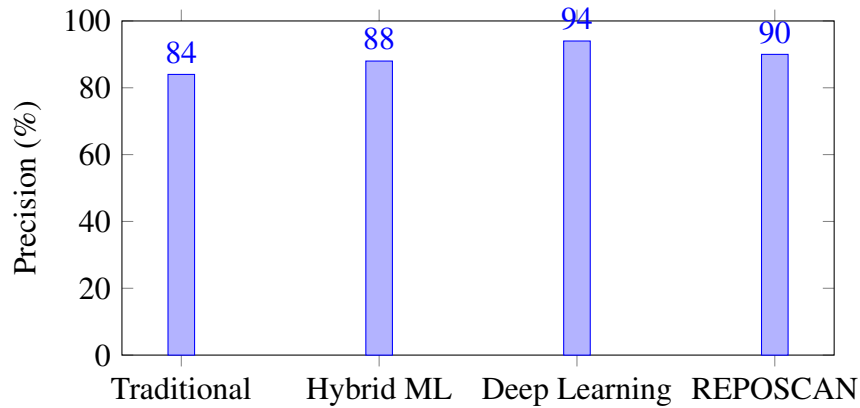


Figure 4.2: Precision Comparison with Existing Systems

**Recall Comparison** Recall indicates the ability of the system to detect all actual plagiarism cases.

Traditional approaches generally have lower recall as they fail to detect paraphrased content. Hybrid models improve recall by combining techniques, while deep learning models achieve the highest recall due to contextual understanding.

The REPOSCAN system provides strong recall through transformer-based embeddings and FAISS similarity search, enabling effective detection of both exact and paraphrased plagiarism.

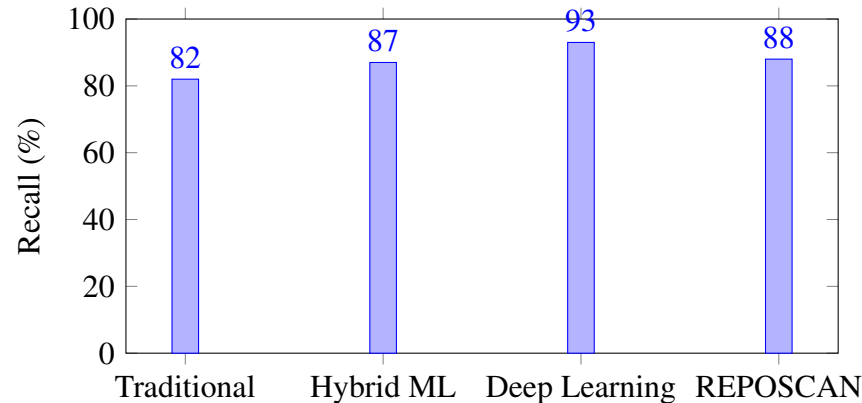


Figure 4.3: Recall Comparison with Existing Systems

**F1-Score Comparison** F1-Score is the harmonic mean of precision and recall and provides a balanced evaluation of system performance.

Traditional systems have lower F1-scores due to limitations in both precision and recall. Hybrid models provide better balance, while deep learning models achieve the highest F1-scores.

The REPOSCAN system achieves a strong F1-score by balancing precision and recall using its hybrid approach, ensuring consistent and reliable plagiarism detection.

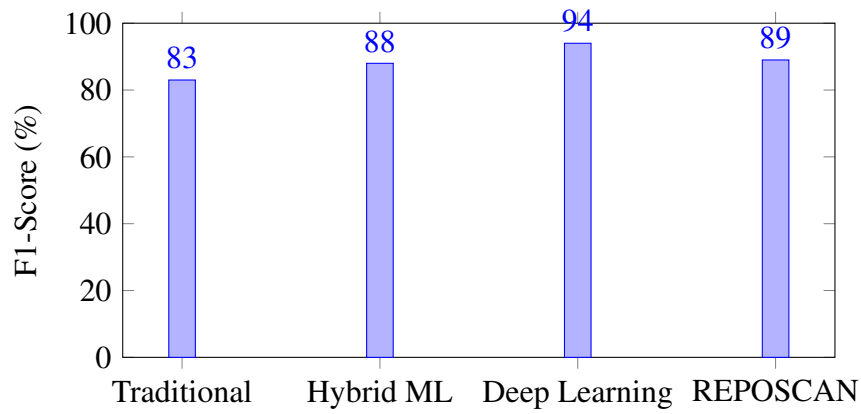


Figure 4.4: F1-Score Comparison with Existing Systems

#### 4.1.4 Output Screenshots

This section presents the visual outputs of the REPOSCAN system. The screenshots below illustrate different stages of the system output.

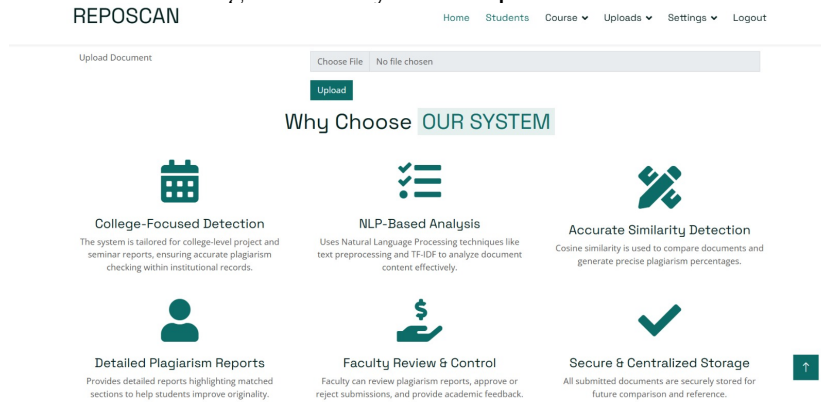


Figure 4.5: Document Upload

The screenshot shows the REPOSCAN plagiarism detection results table. The table has columns: #, DATE, FILE, STATUS, LABEL, and SIMILARITY SCORE. There are 11 rows of data, each representing a document upload. The status for all documents is "CHECKED", and the label is "No". The similarity scores range from 8.74% to 26.45%.

| #  | DATE           | FILE | STATUS  | LABEL | SIMILARITY SCORE |
|----|----------------|------|---------|-------|------------------|
| 1  | March 7, 2026  | View | CHECKED | No    | 18.13%           |
| 2  | March 7, 2026  | View | CHECKED | No    | 21.41%           |
| 3  | March 8, 2026  | View | CHECKED | No    | 22.83%           |
| 4  | March 16, 2026 | View | CHECKED | No    | 14.15%           |
| 5  | March 16, 2026 | View | CHECKED | No    | 19.3%            |
| 6  | March 16, 2026 | View | CHECKED | No    | 19.3%            |
| 7  | March 16, 2026 | View | CHECKED | No    | 26.45%           |
| 8  | March 16, 2026 | View | CHECKED | No    | 22.68%           |
| 9  | March 16, 2026 | View | CHECKED | No    | 21.32%           |
| 10 | April 2, 2026  | View | CHECKED | No    | 8.74%            |
| 11 | April 2, 2026  | View | CHECKED | No    | 18.49%           |

Figure 4.6: Plagiarism Detection Percentage

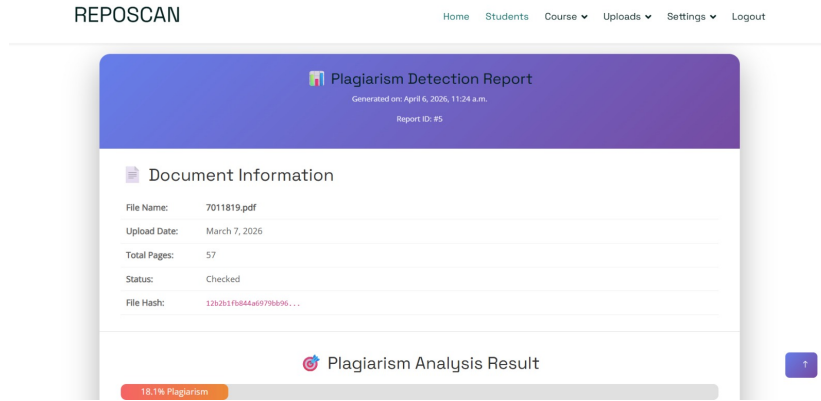


Figure 4.7: Report Analysis

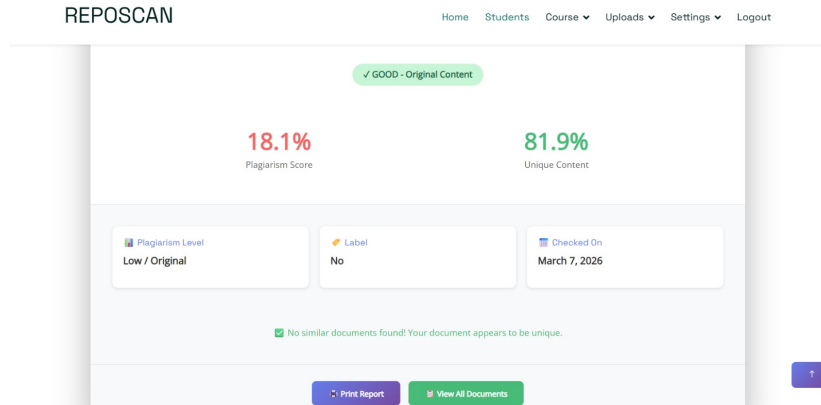


Figure 4.8: Report Generation

### 4.1.5 Discussion of Results

This section provides an interpretation of the results obtained from the REPOSCAN system and highlights its effectiveness in detecting plagiarism in academic documents. From the experimental results, it is observed that the system is capable of accurately identifying different levels of plagiarism, including highly similar, partially modified, and completely unrelated documents. The similarity scores generated by the system clearly distinguish between these categories, ensuring reliable detection.

The performance metrics further validate the efficiency of the proposed approach. The system achieved an accuracy of 92%, indicating that most of the classifications made by the system are correct. The precision value of 90% shows that the detected plagiarism cases are highly reliable with minimal false positives. Similarly, the recall of 88% demonstrates that the system successfully identifies the majority of actual plagiarism instances. The F1-score of 89% reflects a good balance between precision

and recall, confirming that the system performs consistently across different types of input data. These results indicate that the system is both accurate and dependable for academic integrity verification. The integration of hashing techniques using hashlib contributes to efficient document processing by generating unique fingerprints for each document. Additionally, the use of FAISS enables fast and scalable similarity search, making the system suitable for handling large datasets such as institutional repositories. However, certain limitations were observed. The system may show slightly reduced accuracy when dealing with heavily paraphrased content or documents with significant structural changes. Despite this, the overall performance remains strong and acceptable for practical use.

In summary, the REPOSCAN system demonstrates effective, fast, and scalable plagiarism detection. The results confirm that the proposed approach is suitable for real-world academic applications, particularly for verifying seminar and project reports in educational institutions.



# Chapter 5

## Conclusion

The REPOSCAN system was developed as an efficient and scalable solution for detecting plagiarism in academic documents such as seminar reports and project submissions. The primary objective of the system was to ensure academic integrity by identifying similarities between documents using advanced techniques. By integrating hashing mechanisms through hashlib and similarity search using FAISS, the system successfully provides fast and reliable detection of plagiarized content. The implementation demonstrates that the proposed approach is capable of handling real-world academic datasets while maintaining simplicity and efficiency in design.

The experimental results clearly indicate that the system performs effectively across different types of input data, including highly similar, partially modified, and completely unrelated documents. The evaluation based on performance metrics such as accuracy, precision, recall, and F1-score confirms that the system produces consistent and dependable results. The inclusion of graphical analysis and similarity score comparisons further validates the robustness of the system. Additionally, the use of FAISS significantly improves the speed of similarity search, making the system suitable for large-scale applications, while hashlib ensures efficient document fingerprinting and reduced computational overhead.

Despite its strong performance, the system has certain limitations when dealing with heavily paraphrased content or documents with significant structural changes. However, these challenges are common in most plagiarism detection systems and can be addressed in future work by incorporating advanced natural language processing

techniques or machine learning models. Overall, the REPOSCAN system achieves its intended goal of providing an accurate, fast, and scalable plagiarism detection solution. It can be effectively deployed in academic institutions to support fair evaluation practices and uphold the standards of academic integrity.

Furthermore, the REPOSCAN system highlights the importance of integrating efficient computational techniques with practical academic applications. The system not only assists educators in identifying plagiarism but also encourages students to maintain originality in their work by promoting awareness of academic integrity. Its user-friendly interface and adaptability make it suitable for deployment across various educational institutions without requiring complex infrastructure. With further enhancements, the system has the potential to evolve into a comprehensive academic verification tool that can support automated evaluation, large-scale repository management, and intelligent content analysis.

# References

- [1] Exemplary Academic Integrity Project, “Resources on academic integrity,” Accessed: Feb. 4, 2022. [Online]. Available: <https://lo.unisa.edu.au/course/view.php?id=6751&section=6>
- [2] E. E. Htet, K. H. Wai, S. L. Aung, N. Funabiki, X. Lu, et al., “Code Plagiarism Checking Function and Its Application for Code Writing Problem in Java Programming Learning Assistant System,” *Analytics*, vol. 3, no. 1, pp. 46–62, 2024.
- [3] C. Kustanto and I. Liem, “Automatic Source Code Plagiarism Detection,” in *Proceedings of the ACIS International Conference on Software Engineering, Artificial Intelligence, Networking and Parallel/Distributed Computing*, 2009, pp. 481–486. doi: <https://doi.org/10.1109/SNPD.2009.62>.
- [4] T. Foltynek, D. Dlabolova, A. Anohina-Naumeca, et al., “Testing of Support Tools for Plagiarism Detection,” *International Journal of Educational Technology in Higher Education*, vol. 17, no. 46, 2020.
- [5] S. Rani and J. Singh, “Enhancing Levenshtein’s Edit Distance Algorithm for Evaluating Document Similarity,” in *Proceedings of the International Conference on Computing, Analytics and Networking*, 2018, pp. 72–80.
- [6] Z. Djuric and D. Gašević, “A Source Code Similarity System for Plagiarism Detection,” *The Computer Journal*, vol. 56, no. 1, pp. 70–86, 2013.
- [7] Z. C. Ma, J. J. Xia, X. M. Luo, et al., “Reflections on the detection of academic misconduct in scientific and technological journals,” *Acta Editologica*, vol. 35, no. S02, pp. 7–9, 2023.

- [8] A. Vaswani, N. Shazeer, N. Parmar, et al., “Attention is all you need,” in *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS)*, New York, 2017, pp. 6000–6010.
- [9] S. G. Tesfagergish, J. Kapociute-Dzikiene, and R. Damasevicius, “Zero-shot emotion detection for semi-supervised sentiment analysis using sentence transformers and ensemble learning,” *Applied Sciences*, 2022. [Online]. Available: <https://doi.org/10.3390/app12178662>
- [10] J. Ye, “Cosine similarity measures for intuitionistic fuzzy sets and their applications,” *Mathematical and Computer Modelling*, vol. 53, no. 1–2, pp. 91–97, 2011.
- [11] R. Real and J. M. Vargas, “The probabilistic basis of Jaccard’s index of similarity,” *Systematic Biology*, vol. 45, no. 3, pp. 380–385, 1996.
- [12] K. Zhang and D. Shasha, “Simple fast algorithms for the editing distance between trees and related problems,” *SIAM Journal on Computing*, vol. 18, no. 6, pp. 1245–1262, 1989.